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## PRICE

- Date
- Open
- High
- Low
- Close
- Percent change

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## SESSION CONTEXT

- **% Change:** Session return from prior close to current close. Captures both the overnight gap and the intraday move. Positive = Bullish session; negative = Bearish
- **Gap %:** Difference between today's open and yesterday's close, as a percentage. A positive gap signals overnight bullish sentiment; negative signals bearish. Gaps above 0.05% are meaningful for daily FX.
- **Range (pips):**  $(\text{High} - \text{Low}) \times 10,000$ . Total session volatility in pips. A wide range vs recent average indicates a high-conviction session; a narrow range signals consolidation or indecision.
- **Body Fill %:**  $|\text{Close} - \text{Open}| \div (\text{High} - \text{Low}) \times 100$ . What fraction of the total range was captured by the candle body. High fill = directional conviction; low fill = wick-dominated indecision.
- **Inside Bar:** Today's high is below yesterday's high AND today's low is above yesterday's low — the full session is 'inside' the prior bar. Signals volatility contraction. Breakouts above the high or below the low are common entry triggers Highlighted in amber on the chart.

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## VOLUME

- **Volume Bull:** Volume on sessions where  $\text{close} \geq \text{open}$ . Green bars signal buying-side participation. High bullish volume confirms upside moves and indicates institutional demand behind the advance.
- **Volume Bear:** Volume on sessions where  $\text{close} < \text{open}$ . Red bars signal selling-side participation. High bearish volume on declining sessions confirms distribution and downside conviction from larger players.
- **Vol SMA(20):** 20-session simple moving average of volume — the baseline for measuring weather participation is elevated or suppressed. Volume significantly above this line amplifies the significance of the concurrent price move; volume below it suggest low conviction conditions.

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## BOLLINGER

- **Bollinger Band Upper / Lower:** Bollinger bands are set to standard deviations above and below a 20 period moving average. Price, touching or piercing the upper band signals overbought conditions and ranging markets, or breakout strength and trending markets. The lower band signals oversold or bearish momentum. Bands widen and high volatility regimes in contract during squeezes.

- **Bollinger Band Midline (SMA20):** the Bollinger midline is a 20. Simple moving average and act as the mean reversion target in ranging markets. In trending environments it acts as dynamic support (uptrend) or resistance (downtrend). Bollinger band %B measures where the clothes sits within the band: 100% = at upper band, 50% = at midline, 0% = at lower band.
- **Historical Volatility:** historical volatility (Hist Vol) is the annualized standard deviation of log returns — a statistical measure of how much price has dispersed over recent sessions. Rising Hist Vol confirms an active, trending market. Falling Hist Vol signals consolidation. Use alongside Bullinger bandwidth to cross check whether volatility compression is genuine or a data artifact.

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## MACD

- **Histogram:** the difference between the MACD line and the signal line, plotted as bars. Green bars mean MACD is above signal (bullish momentum); red bars mean MACD is below signal (bearish momentum). Growing bars, signal accelerating momentum; shrinking bars, warn of a slow down or reversal. The histogram turns before the lines cross, making it a leading indicator of momentum shifts.
- **MACD Line:** calculated as  $EMA(12) - EMA(26)$ . It measures the difference between two exponential moving averages of price. When MACD is positive, the short term average is above the long-term — a bullish condition. When negative, bearish. Crossovers of the zero line are used as trend, change signals, though they lag price action.
- **Signal Line:** A 9-period EMA of the MACD line itself. It acts as a trigger: when MACD crosses above the signal line, it generates a bi signal; when it crosses below, a sell signal. The signal line smooths the MACD and reduces noise. Crossovers in extreme territory (far from zero) are considered more reliable than those near zero.

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## PIVOT POINTS

- **Resistance 1 / Resistance 2 / Resistance 3:** Resistance levels derived from the prior sessions, high, low, and close. Resistance 1 is the first target above the pivot; resistance 2 and resistance 3 mark progressively stronger resistance. A close above any resistance level, turns it into new support. Resistance 3 is rarely tested in a single session — reaching it signals in exceptionally strong, bullish move.
- **Support 1 / Support 2 / Support 3:** Support levels below the classic pivot point. Support 1 is the first demand zone: support 2 and support 3 are deeper levels that come into play on pronounced selling sessions. A closed below any support level turns it into new resistance. Support 3 violations are typically associated with high volatility, trend driven sessions.

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## KELTNER

- **Upper Band:** the upper Keltner band =  $EMA(20) + 2 \times ATR(10)$ . A close above this band signals, a bullish, breakout or overbought extreme, depending on context. In a strong trend, price can ride the upper band. In a range, it marks a reversal zone.
- **Midline:** the Keltner midline is an  $EMA(20)$  of price. It acts as a dynamic magnet — in ranges price gravitates back to it, and in transit acts as for support (uptrend) or resistance (downtrend). The direction of the midline shows the current trend slope.

- **Lower Band:** the lower Kelner band =  $\text{EMA}(20) - 2 \times \text{ATR}(10)$ . A close below signals bearish momentum, or in oversold extreme. In down trends, price controlled the lower band. In ranges, it marks a potential mean reversion long zone back toward the midline.

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## EMAs

- **EMA 9:** 9-period exponential moving average. The fastest line, highly responsive to recent price action. Access dynamic short term support in up trends and resistance and down trends. They close below EMA9 in an uptrend is often the first warning of weakness.
- **EMA 20:** 20-period exponential moving average. A responsive short-to-medium-term trend filter. Because EMAs weight recent prices more heavily than SMAs, EMA20 reacts faster to price changes. Price holding above the EMA20 in an up trend confirms short-term bull structure; a sustained break below is an early warning of trend weakness.
- **EMA 50:** 50-period exponential moving average. The primary medium-term trend reference. Faster reacting than SMA50, it gives earlier signals on trend changes. Institutional traders watch EMA 50 closely — a price cross above or below this level often triggers significant order flow. Uptrends require price above EMA50.
- **EMA200:** 200-period exponential moving average. The long-term trend anchor. Price above EMA200 = bull market context; below = bear market. The EMA200 golden Cross (EMA50 crossing above EMA200) and the death cross (EMA50 crossing below) are widely followed structural signals — EMA versions react, fast faster than their SMA equivalents.

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## RSI (9)

- **RSI (9):** relative strength index over 9 periods. Measures the speed and magnitude of recent price moves on a 0 - 100 scale. Readings above 70 signal overbought conditions; below 30 signals over sold. The short 9-period look back makes it highly responsive — useful for spotting, short-term exhaustion quickly, but prone to false signals in trending markets.
- **Stochastic RSI %K:** the Stochastic applied to RSI values rather than price. %K is the raw line: it shows where the current RSI sits within its own high-low range over the look back period, scaled 0 - 100. It oscillates far more aggressively than raw RSI and is sensitive to short-term momentum shifts. Values above 80 = overbought; below 20 = oversold.
- **Stochastic RSI %D:** A smoothed 3-period moving average of %K. Because %D lags %K slightly, crossovers between the two are used as signals: %K crossing above %D is a buy signal; crossing below is a sell signal. %D filters out some of the noise in the raw %K line, making crossovers more reliable when both lines are in extreme territory.

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## RSI (14)

- **RSI (14):** relative strength index over 14 periods — the standard setting. Smoother and more reliable than RSI(9), it filter filters out short-term noise and gives fewer but higher quality signals. Above 70 = overbought; below 30 = oversold. Because it uses a longer look back, overbought/oversold readings here carry more weight than on the 9-period.
- **Trend (5-bar SMA of RSI):** A 5-period simple moving average applied to RSI(14) itself. When RSI(14) is above the Trend line, momentum is accelerating upward; when below, momentum is fading. The crossover between RSI(14) and its Trend line is an early signal of a momentum shift — useful for timing entries and exits within the broader RSI context.

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## ADX

- **+DI (14):** the positive directional indicator measures the strength of upward price movement over 14 periods. When +DI is above -DI, bulls are in control of directional movement. A rising +DI confirms strengthening, upside pressure; a falling +DI warns that bullish momentum is fading.
- **-DI (14):** the negative directional indicator measure measures the strength of downward price movement. When -DI is above +DI, bears dominate. A rising -DI signals increasing selling pressure. A crossover of -DI above +DI with ADX above 25 is a classic bearish trend entry signal.
- **ADX:** the average directional index measures trend strength, not direction — it rises in both up and down trends. ADX below 20 = ranging market (avoid trend signals). ADX above 25 = trending. ADX above 40 = strong trend. A rising ADX confirms a trend is developing; a falling ADX signals the trend is fading regardless of direction.

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## ICHIMOKU

- **Tenkan:** conversion line:  $(9\text{-period high} + \text{low}) \div 2$ . Plotted at the current bar. Short-term momentum — a TK cross above Kijun is a bullish entry signal. Acts as near-term support in up trends.
- **Kijun:** Base Line:  $(26\text{-period high} + \text{Low}) \div 2$ . Plotted at the current bar. Medium term trend anchor, and the most important Ichioku S/R level. Price above Kijun = bullish; below = bearish.
- **Senkou A:** Leading Span A:  $(\text{Tenkan} + \text{Kijun}) \div 2$ , plotted 26 bars FORWARD. Forms the faster cloud boundary. When  $A > B$  the cloud is bullish. The projected cloud ahead shows anticipated support and resistance.
- **Senior B:** Leading Span B:  $(52\text{-period high} + \text{low}) \div 2$ , plotted 26 bars FORWARD. The slower, stronger cloud boundary. When  $B > A$  the cloud is bearish. Thicker cloud = stronger S/R zone. Projected cloud gives a look 26 bars ahead.
- **Chikou:** Lagging span: today's close plotted 26 bars BACK. Confirms trend by comparing current clothes to historical price. Chico above price from 26 bars ago = bullish; below = bearish. It stops 26 bars before the current bar since future data is unavailable.

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## ATR

- **ATR(14):** average true range of 14 periods. True range is the greatest of: current High-Low, | High-Prior Close|. ATR averages these over 14 bars, expressed in pips for Forex. Rising ATR = expanding daily ranges, active market, wider stops required. Falling ATR = compression, quieter conditions, tighter stops viable. Use a 1-1.5x ATR multiplier to set stops outside the noise of normal daily movement.
- **ATR SMA(20):** A 20-bar simple moving average of ATR(14). Serves as the benchmark for 'normal' volatility. When ATR is significantly above its SMA20 ( $>25\%$ ), the market is in an elevated-volatility regime — size down. When ATR significantly below it's SMA20 ( $<75\%$ ), the market is compressing — a precursor of volatility, breakout, and sharp directional move. The SMA20 is the reference line that contextualizes whether today's ATR is a warning or routine.

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## CANDLE CONTEXT

- **Tier 1 — Multi-Candle Story:** these patterns require a prior directional move plus a pivot or pause candle, then a follow-through confirmation. They tell a complete story across 2-3 sessions: a direction, and exhaustion, and a rehearsal. Morning/Evening Star, Engulfing variants, Piercing Line, and Dark Cloud Cover all belong here. Highest reliability — the market is explicitly showing a change of control.
- **Tier 2 — Single Candle Rejection:** single candles with an extreme wick showing decisive day rejection of a price level. The key factor is context: a hammer at the bottom of a decline is bullish, the same shape after a rally (hanging man) is bearish. Inverted hammer needs a bullish confirmation candle; shooting star needs a bearish follow-through. The candle shaped alone is not enough — location, and next session matter.
- **Tier 3 — Needs Confirmation:** solid, structural signals that require the next candle to confirm. Tweezers show a level being defended twice in consecutive sessions — a tested support resistance. Harami patterns, show momentum, stalling, inside the prior candles, body, and the Harami Cross version uses a dojo for maximum indecision. These are early alerts, not entry signals — always wait for the confirming session before acting.

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## FAILURE SWING

- **Failure Swing CDF:** the cumulative distribution function plots the probability that a failure swing is  $\leq$  a given pip value. Find any pip value on the x-axis, trace up to the blue line, then read the y-axis — that percentage of qualifying sessions had a failure swing at or below that level. A steep early rise means most failure swings are small and clustered near zero. A gradual slope indicates a wide distribution — price regularly makes large intraday wicks before reversing.
- **Percentile Bands:** dash lines mark the 25th, 50th (median), 75th, and 90th percentiles. Is 75th percentile, is the most actionable reference: it separates routine intraday noise from meaningful rejection when the yellow dateline sits above the 70th percentile, the market is showing above average intraday candle rejection. Above the 90th marks and extreme session. Below the 50th is a quiet, low conviction day with little directional conviction inside the candle.

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## MARKET STRUCTURE

- **Swing High (SH):** a local price peak confirmed by being the highest high within the surrounding N bars on each side. Mark's potential resistance and defines the upper boundary of structure. Used to identify HH/LL sequences.
- **Swing Low (SL):** a local price trough confirmed by being the lowest low within the surrounding N bars on each side. Mark's potential support and the lower boundary of structure. Used to identify HL/LL sequences.
- **BOS (Break of Structure):** price closes beyond the last swing high (bull BOS) or swing low (bear BOS) in the direction of the prevailing trend, confirming continuation. Smart money is defending and extending the move.

- **CHOCH (Change of Character):** price breaks a swing level counter to the current trend, signaling a potential reversal. CHOCH is the first warning that dominant structure is failing. Watch for follow-through confirmation before acting.

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## WILLIAMS %R

- **Williams %R:** measures where the most recent close sits within the high-low range of the last 14 bars. Ranges from -100 (close at the period low—most oversold) to 0 (close at the period high — most overbought). A fast, reactive oscillator that leads price turns.
- **Overbought >-20:** When %R rises above -20 the close is in the top 20% of the recent range. In a trending market, this confirms bullish strength. In a ranging market or after an extended rally it signals reversal risk — watch for %R to roll back below -20 as an exit trigger.
- **Oversold <-80:** When %R drops below -80 the close is in the bottom 20% of the recent range. Potential mean reversion bounce zone. Confirmation requires %R crossing back above -80. In a strong downtrend oversold can persist — don't buy the first touch; wait for the cross.

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## CCI

- **CCI:** Commodity channel index — measures how far price has deviated from its statistical mean. A leading oscillator: it turns before price and excels at identifying momentum, extremes, and divergences.
- **CCI MA(14):** 14-period simple moving average of CCI. Smooths the raw CCI line so shorter term noise doesn't obscure the underlying momentum cycle. Crossovers between CCI and its MA often signal early momentum shifts before the raw line reaches an extreme.
- **Levels:** Above +100 = overbought (bullish momentum extreme — consider exits or tightened stops). Below -100 = oversold (bearish extreme — watch for bounce).  $\pm 100$  to 0 = neutral range. Above +200 or below -200 = unsustainable extreme — high mean-reversion probability.

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## RATE OF CHANGE (5)

- **ROC(5):** Rate of change over five sessions:  $(\text{Close} - \text{Close}[5]) \div \text{Close}[5] \times 100$ . Measures the net percentage gain or loss of exactly one trading week. Positive = net bullish week; negative = net bearish. Accelerating ROC in the direction of the trend confirms momentum; decelerating ROC warns of fatigue. Extreme readings signal overextension.
- **Above Zero:** ROC above zero means this week's close is higher than the close five sessions ago — net buying pressure over the look back window. The magnitude matters: readings above +0.5% indicate meaningful upside momentum. Readings approaching the 20 session peak rate suggest the move is maturing.
- **Below Zero:** ROC below zero means this week's close is lower than the close five sessions ago — net selling pressure. Readings below -0.5% indicate meaningful

downside momentum. Watch for ROC divergence: if price makes a new low, but ROC fails to confirm, bearish momentum may be fading.